Thermodynamic saturation and window barriers
for graded spaces in a subexponential 3-SAT algorithm

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Abstract

Motivated by the graded four-leg construction in the subexponential 3-SAT algorithm proposed by Anthropic, we study the normalized costs of prescribed graded windows, rather than the intrinsic ranks of the resulting factorization matrices. For every compact interval $\mathcal{J} = [x_1, x_2] \Subset (0, \infty)$, we establish for Kleinian-code graded spaces V_S the uniform estimate

$$\rho_{\mathrm{win}}(V_S; h, \lambda) \geq \frac{1}{\sqrt{2}} + \eta_{\mathcal{J}} - C_{\mathcal{J}} m^{-1/2}, \qquad h/m \in \mathcal{J}, \quad 2^{\lambda} \leq \dim(V_S)_h,$$

where $\eta_{\mathcal{J}} > 0$ and the constants do not depend on the isotropic additive code $S \subset \mathbb{K}^m$. The proof combines a central-window bound, a sectorwise thermodynamic sign criterion and certified interval arithmetic. No concavity or asymptotic convergence of the code-weight spectrum is assumed. In particular, this fixed-density bound does not resolve the single-block regime $h/m \to \infty$ or determine intrinsic factorization ranks.

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1 Introduction

1.1 Context and motivating problem

The Exponential Time Hypothesis (ETH) of Impagliazzo and Paturi [1] asserts that 3-SAT does not admit a worst-case $2^{o(n)}$-time algorithm; the sparse-formula perspective and related reductions were developed by Impagliazzo, Paturi and Zane [2]. In October 2026, OpenAI and Anthropic released proposed subexponential counting algorithms for SAT [3, 4]. Their mathematical status and broader consequences are independent of the narrower question investigated here.

Anthropic's Appendix B studies a four-leg representation associated with the *crossing tensor*

$$X_\lambda(x, y; u, v) = [x = v][y = u], \qquad x, y, u, v \in \{0, 1\}^\lambda, \tag{1.1}$$

where λ is the number of encoding bits. The construction is organized using a graded vector space with a selected encoding degree $h > 0$; its prescribed intermediate windows determine the two factorization-dimension budgets. In the original partition example, these dimensions are governed by partition numbers $p(E)$, whose asymptotic growth is classical [8], and the normalized window cost approaches $1/\sqrt{2}$. Whether other graded spaces can produce less expensive windows is a natural dimension-level question, but the existence of an algebraic factorization is a further requirement.

For a graded space V, write $D(E) = \log_2 \dim V_{\le E}$ for the cumulative dimension profile. The window-cost parameter ρ_{win}, defined precisely in Section 2, measures the largest pair of logarithmic window dimensions normalized by 4λ. The present work concerns families V_S built from additive isotropic codes $S \subset \mathbb{K}^m$ over the Klein four-group, where m is the block number.

1.2 Main result and proof mechanism

Theorem 1.1 (Main result: strict fixed-density window barrier)**.** *Fix a compact interval $\mathcal{J} = [x_1, x_2] \Subset (0, \infty)$. There exist constants $\eta_{\mathcal{J}} > 0$ and $C_{\mathcal{J}} > 0$, depending only on $\mathcal{J}$ and the fixed character alphabet, such that for every $m \ge 1$, every additive isotropic code $S \subset \mathbb{K}^m$, every admissible positive degree h with $h/m \in \mathcal{J}$, and every $\lambda \ge 1$ satisfying $2^\lambda \le \dim(V_S)_h$, one has*

$$\rho_{\text{win}}(V_S; h, \lambda) \;\ge\; \frac{1}{\sqrt{2}} + \eta_{\mathcal{J}} - \frac{C_{\mathcal{J}}}{\sqrt{m}}. \tag{1.2}$$

Theorem 6.5 gives the proof. In particular, the prescribed window cost is strictly above $1/\sqrt{2}$ for all sufficiently large m at each fixed compact density interval.

The proof has three ingredients. First, the single central window $s = 0$ gives $\rho_{\text{win}} \ge D(2h)/(2\lambda)$ (Lemma 2.4). Second, a thermodynamic scaling principle converts monotonicity of $\beta \log Z(\beta)$ into lower bounds on graded dimension growth (Lemma 4.2). Third, a strictly negative *sectorwise* derivative bound for the two D_4 coset characters holds uniformly across all isotropic codes (Theorem 6.4 and Appendix A); its numerical sign subchecks are independently implemented in Arb and `mpmath.iv`. The strict margin in (1.2) is a fixed-compact consequence of this derivative inequality.

1.3 Relation to previous work

Brito [5] derives encoding/rank obstructions for Appendix-B piece families, including the $1/\sqrt{2}$ threshold in the class of exact power-law growth profiles. Majdoddin [6] examines more general dimension-profile shapes, obtains a 2/3 lower bound under his concavity and window hypotheses, and discusses the need

for nonstandard growth profiles to improve the partition-based shape. Tulsyan [7] studies rank/spread obstructions for specified graded constructions, with machine-checked components and a $1/\sqrt{2}$-scale conclusion under the hypotheses stated there. These overlapping barriers are important predecessors, not results claimed as original in this paper.

Work	Setting / assumptions	Relevant conclusion
Brito [5]	Graded pieces of the Appendix-B shape; exact power-type profiles and encoding/rank constraints.	A $1/\sqrt{2}$ floor for the analyzed power-profile route.
Majdoddin [6]	Dimension profiles of four-leg pieces; concavity and stated window-size conditions.	A 2/3 lower bound for concave profiles in the specified regime.
Tulsyan [7]	Graded two-order slices subject to spread and pairing conditions.	A $1/\sqrt{2}$-scale obstruction for the stated graded models.
This paper	Isotropic Kleinian codes, $h/m \in \mathcal{J} \Subset (0,\infty)$; no convergence or concavity assumption on code weight spectra.	The strict *window-dimension* bound $1/\sqrt{2}+\eta_{\mathcal{J}} - O_{\mathcal{J}}(m^{-1/2})$.

Table 1: Comparison of hypotheses and conclusions. The rows compare stated results; they do not establish logical non-implication or independent priority.

The distinguishing feature of Theorem 6.5 is its uniformity in the varying block number and isotropic code, together with a finite-size error and a strictly positive margin on every fixed compact density interval. This is a precisely delimited statement; its priority relative to possible consequences of the cited literature requires mathematical comparison beyond matching the published theorem statements.

1.4 Scope and limitations

The inequality concerns *prescribed intermediate window dimensions*, not the intrinsic flattening ranks of any particular factorization matrix. These are different quantities; indeed, $\rho_{\mathrm{rank}} \le \rho_{\mathrm{win}}$ in the framework of Lemma 2.2. No Appendix-B-compatible operators or new four-leg identities are constructed for V_S, and the result therefore supplies neither a new SAT algorithm nor a universal obstruction to more efficient pieces. Its positive gap is uniform only for fixed compact h/m intervals, not in the high-density regime $h/m \to \infty$ (or the low-density regime $h/m \to 0$). The numerical certificates verify specified sign inequalities subject to the stated analytic reductions; an independent human expert review of the whole argument has not yet been completed.

2 Window factorizations

A graded space $V = \bigoplus_{E\in\Gamma} V_E$ has discrete $\Gamma \subset [0,\infty)$ and finite-dimensional components. Put

$$V_{\le E} = \bigoplus_{e\le E} V_e, \quad D(E) = \log_2 \dim V_{\le E}, \quad V_{[a,b]} = \bigoplus_{e\in[a,b]} V_e, \quad W[a,b] = \log_2 \max\{1, \dim V_{[a,b]}\}.$$

Definition 2.1 (Window cost)**.** *For $h > 0$, $\lambda \ge 1$ and $|s| \le 2h$ set*

$$C_h(s) = W[s, s+2h] + W[-s, 2h-s], \qquad \rho_{\mathrm{win}}(V; h, \lambda) = \frac{1}{4\lambda} \sup_{|s|\le 2h} C_h(s).$$

Exact dictionary with Anthropic. In [4, Appendix B, equations (54)–(56)], $V = \mathbb{Q}[t_1, t_2, \ldots]$ has integer weights, $\dim V_E = p(E)$, and the 2^λ inputs are encoded by distinct monomials in weight h (chosen as the least integer with $p(h) \ge 2^\lambda$). The pieces are indexed by $I = \{-2h, -2h+1, \ldots, 2h\}$, including $s = 0$. In Definition 2.1 we allow real s, while the source uses integer s; we use logarithms of window dimensions, whereas the source takes the ceilings of the binary logarithms. We also allow any λ satisfying the encoding condition, whereas the source chooses the *least integer* h for which $p(h) \ge 2^\lambda$. These differences must be maintained when comparing costs; all arguments below use the full real window range and the stated grading. The row and column factorization dimensions of piece s are, respectively,

$$R_s^{\mathrm{row}} = \sum_{i=0}^{2h} p(s+i), \qquad R_s^{\mathrm{col}} = \sum_{i=0}^{2h} p(2h-s-i),$$

where $p(j) = 0$ for $j < 0$, and their bit costs are the ceilings of the binary logarithms. In [4, Definition 7], the parameters satisfy $1/2 \le \rho < 1$ and $c \ge 1$ is an integer. The *two-direction bit-budget condition* is specifically Definition 7(b):

$$c_s^{\mathrm{row}} + c_s^{\mathrm{col}} \le 4\rho\lambda + c\log(2\lambda). \tag{2.1}$$

Separately, Definition 7(c) requires

$$|I| \le (2\lambda)^c,$$

and imposes magnitude bounds on the indices, coefficients and factorization data, together with effective-computability requirements. Definition 7(a) is the crossing-identity requirement. Thus (2.1) is only the bit-budget portion of the source definition, not a claim that a new graded space provides all admissible pieces. The $4\rho\lambda+8$ estimate in the source Lemma 2(c) concerns its main-text $\rho = 11/12$ pieces and should *not* replace Definition 7(b) in the Appendix-B comparison. The bound on ρ_{win} transfers to the specified window-cost form of (2.1), with the logarithmic tolerance, in Corollary 6.12. Our construction of graded spaces alone supplies no operators satisfying the source identities.

A family of matrices $\{A_s\}_{s\in I}$ summing to dX_λ ($d \ne 0$) is a window factorization over V when each row flattening factors through $V_{[s,s+2h]}$ and each column flattening through $V_{[-s,2h-s]}$.

Lemma 2.2 (Ranks versus windows)**.** *For such a factorization,*

$$\rho_{\mathrm{rank}} := \frac{1}{4\lambda}\max_s\bigl(\log_2 \mathrm{rank}_{\mathrm{row}} A_s + \log_2 \mathrm{rank}_{\mathrm{col}} A_s\bigr) \le \rho_{\mathrm{win}}.$$

The reverse inequality need not hold.

Proof. The rank of a factorization through an N-dimensional space is at most N. □

The source encoding gives the constraint

$$2^\lambda \le \dim V_h, \qquad \lambda \le D(h) \qquad \text{(E)}$$

Lemma 2.3 (Rank floor)**.** *If dX_λ is the sum of $|I|$ such pieces, then*

$$2\lambda \le D(4h) + \log_2 |I| \qquad \text{(R)}$$

Proof. Put $N = 2^\lambda$. The row-wise flattening separates (x, y) from (u, v), and the column-wise flattening separates (x, u) from (y, v). Since $X_\lambda(x, y; u, v) = [x = v][y = u]$, both are $N^2 \times N^2$ permutation matrices and therefore have rank $N^2 = 2^{2\lambda}$. Rank subadditivity in the row-wise flattening gives $2^{2\lambda} \le \sum_s \mathrm{rank}_{\mathrm{row}} A_s$. For $s \in [-2h, 2h]$, the stated row and column windows $[s, s + 2h]$ and $[-s, 2h - s]$ lie inside $[-2h, 4h]$. Since the grading is nonnegative, each intermediate window has dimension at most $\dim V_{\le 4h}$. Thus $2^{2\lambda} \le |I| \dim V_{\le 4h}$. The column-wise flattening yields the analogous estimate. □

Lemma 2.4 (Central window)**.** *The definition over the* full *range $|s| \le 2h$ gives*

$$\rho_{\mathrm{win}}(V; h, \lambda) \ge \frac{D(2h)}{2\lambda}. \tag{2.2}$$

Under (E) this implies $\rho_{\mathrm{win}} \ge D(2h)/(2D(h))$; under (R) it implies $\rho_{\mathrm{win}} \ge D(2h)/(D(4h) + \log_2 |I|)$.

Proof. At $s = 0$ both windows are $V_{\le 2h}$, so $C_h(0) = 2D(2h)$. □

Role of the central piece. Theorems 4.6 and 6.5 bound the prescribed window cost by estimating this one term of the supremum. Their arguments require the encoding inequality (E), but not the separate rank floor (R). In particular, (2.2) gives no lower bound on the intrinsic ranks of the actual factorization matrices: by Lemma 2.2, those ranks can be smaller than the intermediate-window dimensions.

3 Dimension-only bounds and their limits

Set $\ell = \log_2 |I|/\lambda$. In the original construction $|I| = 4h + 1$ and $\ell \to 0$.

Proposition 3.1 (Profile bounds)**.** *Assume (E) and (R).*

(i) *If* $D(2h) \geq \frac{2}{3}D(h) + \frac{1}{3}D(4h)$ *(in particular for a concave extension on* $[h, 4h]$*), then* $\rho_{\text{win}} \geq \frac{2}{3} - \ell/6$.

(ii) *If* $\chi_h = D(2h)^2/(D(h)D(4h))$*, then* $\rho_{\text{win}} \geq \sqrt{\chi_h/2}\sqrt{1 - \ell/2}$.

(iii) *If* $D(E) = cE^\alpha(1 + \delta_E)$ *for* $E = h, 2h, 4h$*,* $|\delta_E| \leq \delta_0 < 1$*, then*

$$\rho_{\text{win}} \geq \frac{1 - \delta_0}{1 + \delta_0} \max\{2^{\alpha-1},\ 2^{-\alpha}(1 - \ell/2)\} \geq \frac{1 - \delta_0}{1 + \delta_0}\sqrt{\frac{1 - \ell/2}{2}} \qquad (0 \leq \ell < 2).$$

Proof. By (E) and (R), $D(h) \geq \lambda$ and $D(4h) \geq (2 - \ell)\lambda$. Insert each hypothesis into Lemma 2.4. For (ii), $D(2h)^2 = \chi_h D(h)D(4h) \geq \chi_h(2 - \ell)\lambda^2$. For (iii), compare separately $D(2h)/D(h)$ and $D(2h)/D(4h)$. For $0 \leq \ell < 2$, the product of the two branches $2^{\alpha-1}$ and $2^{-\alpha}(1 - \ell/2)$ is $(1 - \ell/2)/2$, so their maximum is at least $\sqrt{(1 - \ell/2)/2}$. The exponent $\alpha = 1/2$ minimizes the uncorrected ($\ell = 0$) maximum. When $\ell \geq 2$, the asserted positive square-root refinement is not used. □

Attribution. The concave-profile obstruction in part (i) and the exact-power-profile obstruction underlying part (iii) are already present in Majdoddin [6, Section 4.7, Proposition 4.24 and subsequent discussion]; see also Brito [5] for power-profile rank bounds. The explicit $\ell = \log_2 |I|/\lambda$ and δ_0 terms are recorded here as elementary quantitative variants, without claiming priority. Part (ii) is likewise included as an elementary comparison rather than a novelty claim. The next proposition addresses dimension-only examples, not realizable operator identities.

Proposition 3.2 (Dimension-only countermodels)**.** (i) *Let* $\dim V_0 = 1$*,* $\dim V_h = 2^\lambda$*,* $\dim V_{4h} = 2^{2\lambda}$*, and all other components vanish. Then (E) and the numerical rank inequality (R) hold, but*

$$\rho_{\text{win}}(V; h, \lambda) = \frac{\log_2(1 + 2^\lambda)}{2\lambda} = \frac{1}{2} + O(2^{-\lambda}/\lambda).$$

(ii) *For a relaxed profile* $D(E) = a + cE^p$ *on* $[h, 4h]$ *(*$p > 0$*),* $t = a/(ch^p) > -1$*, the two constraints alone imply*

$$\rho_{\text{win}} \geq (1 - \ell/2)F_p(t), \qquad F_p(t) = \max\left\{\frac{t + 2^p}{2(t + 1)}, \frac{t + 2^p}{t + 4^p}\right\}.$$

The minimum over t *occurs at* $t = 4^p - 2$*, with value* $\frac{1}{2} + \frac{1}{2(2^p+1)}$*, which tends to* $1/2$ *as* $p \to \infty$.

Proof. For (i), windows with $|s| \leq h$ contain at most $V_0 \oplus V_h$ in each direction; at $s = 0$ both attain that dimension, whereas at an endpoint $s = \pm 2h$ one of the windows contains V_{4h} and the other is the vacuum, giving cost 2λ in the numerator. The stated central cost dominates. For (ii) the first displayed branch decreases with t and the second increases; equality gives $t = 4^p - 2$. □

These are numerical Hilbert-space witnesses: they do not supply the operators or computational certificates required by the source algorithm.

Why impose thermodynamic structure? The countermodels in Proposition 3.2 separate a dimension-profile possibility from an actual algebraic construction: the numerical constraints (E) and (R) alone leave room for a sharp change of growth. The next sections investigate more constrained families using partition functions and graded character identities. In particular, Kleinian glue codes allow weight distributions that need not be concave or convergent across the family; the sectorwise argument in Section 6 still yields the fixed-density window barrier. This is a statement about counted intermediate dimensions, not about the existence or optimality of operator factorization identities.

4 A thermodynamic saturation criterion

4.1 Legendre envelopes and scaling

Let M be graded, with character $Z_M(\beta) = \sum_E (\dim M_E) e^{-\beta E}$ finite for $\beta > 0$. For an arbitrary normalization $m \geq 1$, define

$$\varphi_M(\beta) = \frac{1}{m} \log Z_M(\beta), \qquad \mathcal{L}_M(y) = \inf_{\beta>0} \big[\varphi_M(\beta) + \beta y\big] \in [-\infty, \infty). \tag{4.1}$$

The exponential generating-function (Chernoff) bound, in the language of statistical mechanics and large deviations [14], gives

$$\log \dim M_{\leq my} \leq m \mathcal{L}_M(y). \tag{4.2}$$

Remark 4.1 (Gibbs entropy and logarithmic slope). For a differentiable positive character $Z_M(\beta)$ define its mean energy and Gibbs entropy (in natural-log units) by

$$U_M(\beta) = -\frac{\mathrm{d}}{\mathrm{d}\beta} \log Z_M(\beta), \qquad \mathsf{S}_M(\beta) = \log Z_M(\beta) + \beta U_M(\beta).$$

A direct differentiation gives the identity

$$\frac{\mathrm{d}}{\mathrm{d}\beta} \big[\beta \log Z_M(\beta)\big] = \mathsf{S}_M(\beta) - 2\beta U_M(\beta). \tag{4.3}$$

Consequently, for $\mathsf{S}_M(\beta) > 0$, the monotonicity condition $(\beta \log Z_M)' \leq 0$ is *equivalent* to $\beta U_M / \mathsf{S}_M \geq 1/2$. If the Legendre infimum in (4.1) is attained smoothly at β with $U_M(\beta) = my$ and $\mathcal{L}_M(y) > 0$, then

$$m\mathcal{L}_M(y) = \mathsf{S}_M(\beta), \qquad \frac{\mathrm{d} \log \mathcal{L}_M(y)}{\mathrm{d} \log y} = \frac{y\mathcal{L}_M'(y)}{\mathcal{L}_M(y)} = \frac{\beta U_M(\beta)}{\mathsf{S}_M(\beta)} \geq \frac{1}{2}.$$

Thus the criterion bounds the logarithmic entropy-growth slope from below. Since $\varphi_M = m^{-1} \log Z_M$, its sign condition $(\beta\varphi_M)' \leq 0$ is the same. This is an interpretation under the stated differentiability and positivity assumptions, *not* an extra hypothesis for the next scaling lemma, which requires neither differentiability nor a minimizing inverse temperature.

Lemma 4.2 (Scaling lemma). *If $\psi(\beta) := \beta\varphi(\beta)$ is nonincreasing and $\mathcal{L}(y) = \inf_{\beta>0}[\varphi(\beta) + \beta y]$, then for $c \geq 1$ and $y \in \mathbb{R}$,*

$$\mathcal{L}(cy) \geq \sqrt{c}\, \mathcal{L}(y). \tag{4.4}$$

This includes extended-real infima.

Proof. Set $\beta = \gamma/\sqrt{c}$. Then

$$\mathcal{L}(cy) = \sqrt{c} \inf_{\gamma>0} \left[\frac{\psi(\gamma/\sqrt{c})}{\gamma} + \gamma y\right] \geq \sqrt{c} \inf_{\gamma>0} \left[\frac{\psi(\gamma)}{\gamma} + \gamma y\right] = \sqrt{c}\, \mathcal{L}(y).$$

□

4.2 Product sectors and a finite-length estimate

Definition 4.3 (Block alphabet and sector). *A block alphabet is a finite family $\mathcal{A} = (Z^{(1)}, \ldots, Z^{(r)})$ of characters having nonnegative integral coefficients, finite values for $\beta > 0$, infinitely many nonzero coefficients, and lowest energies $g_j \geq 0$. A product sector M of length m has character*

$$Z_M(\beta) = a_M \prod_{i=1}^{m} Z^{(j_i)}(\beta), \qquad a_M \in \mathbb{Z}_{\geq 1}, \qquad g_M = \sum_i g_{j_i}.$$

Only graded dimensions are used: M need not be a literal tensor product.

Lemma 4.4 (Finite-length Chernoff–Chebyshev). *Let $\mathcal{A}$ be a block alphabet and let $L \geq 0$, $0 < y_1 \leq y_2 < \infty$ and $\delta > 0$. There is a constant $C = C(\mathcal{A}, L, y_2, \delta)$ such that for every length-m product sector M with $a_M \leq e^{Lm}$ and every $y \in [y_1, y_2]$ satisfying $y - g_M/m \geq \delta$,*

$$m\mathcal{L}_M(y) - C\sqrt{m} \leq \log \dim M_{\leq my} \leq m\mathcal{L}_M(y). \tag{4.5}$$

Proof. The upper bound is Chernoff. Put $\mu_j(\beta) = -(\log Z^{(j)})'$ and $v_j(\beta) = (\log Z^{(j)})''$. These are the mean and variance of the canonical energy; μ_j is continuous decreasing, $\mu_j(0+) = \infty$ and $\mu_j(\infty) = g_j$. The first limit follows from the infinitely many nonzero coefficients, with $Z^{(j)}(\beta) \to \infty$ as $\beta \downarrow 0$.

Choose $0 < \beta_{\mathrm{lo}} < \beta_{\mathrm{hi}} < \infty$ such that $\mu_j(\beta_{\mathrm{lo}}) > y_2$ and $\mu_j(\beta_{\mathrm{hi}}) < g_j + \delta/4$ for all j. Put $V_* := \max_{j,\beta\in[\beta_{\mathrm{lo}},\beta_{\mathrm{hi}}]} v_j(\beta) < \infty$. The sector Gibbs distribution is a sum of m independent block energies; its mean is $\mu_m(\beta) = \sum_i \mu_{j_i}(\beta)$ and its standard deviation $\sigma_m(\beta) \le \sqrt{mV_*}$.

Assume $m > 64V_*/(9\delta^2)$. The continuous function $f(\beta) = \mu_m(\beta) + 2\sigma_m(\beta)$ lies above my at β_{lo} and below my at β_{hi}. Indeed $f(\beta_{\mathrm{lo}}) > my_2 \ge my$, whereas $f(\beta_{\mathrm{hi}}) < g_M + m\delta/4 + 2\sqrt{mV_*} < my$. Hence there is $\beta_1 \in [\beta_{\mathrm{lo}}, \beta_{\mathrm{hi}}]$ with $f(\beta_1) = my$. Chebyshev puts at least $3/4$ of Gibbs mass in $[my - 4\sigma_m(\beta_1), my]$. On this interval the weights $e^{-\beta_1 E}$ are at most $e^{-\beta_1(my - 4\sigma_m(\beta_1))}$, so

$$\dim M_{\le my} \ge \frac{3}{4} e^{\beta_1(my - 4\sigma_m(\beta_1))} Z_M(\beta_1).$$

In particular, the explicit intermediate estimate is

$$\log \dim M_{\le my} \ge m\mathcal{L}_M(y) - 4\beta_1\sigma_m(\beta_1) - \log(4/3) \ge m\mathcal{L}_M(y) - 4\beta_{\mathrm{hi}}\sqrt{mV_*} - \log(4/3). \tag{4.6}$$

Finitely many smaller m are handled by the sector ground state and $a_M \le e^{Lm}$, absorbing the bounded range into C. The upper endpoint β_{hi} can be chosen using only the gap δ and the finite alphabet; the constant does not require a separate dependence on y_1 once $y - g_M/m \ge \delta$ is imposed. □

4.3 Assembly

Proposition 4.5 (Sector assembly)**.** *Let $\mathcal{A}$ contain a vacuum character $Z^{(j_0)}$ with $g_{j_0} = 0$. Fix $L, c \ge 0$ and $0 < x_1 \le x_2 < \infty$. There is C' depending only on these data such that if $V = \bigoplus_{\sigma\in\Sigma} M_\sigma$ is a sum of length-m product sectors satisfying*

1. *$|\Sigma| \le (m+1)^c$ and each multiplicity $a_{M_\sigma} \le e^{Lm}$;*
2. *for every sector, $\beta \mapsto \beta\varphi_{M_\sigma}(\beta)$ is nonincreasing;*
3. *some sector has character $(Z^{(j_0)})^m$;*

then for every $h/m \in [x_1, x_2]$ and every $\lambda \ge 1$ with $2^\lambda \le \dim V_h$,

$$\rho_{\mathrm{win}}(V; h, \lambda) \ge \frac{1}{\sqrt{2}} - C' m^{-1/2}. \tag{4.7}$$

Proof. Write $x = h/m$, $\mathcal{L}_\sigma = \mathcal{L}_{M_\sigma}$, and $b = \max_\sigma \mathcal{L}_\sigma(x)$. The vacuum sector gives $b \ge b_0(x_1) > 0$, where $b_0(y) = \inf_{\beta>0}[\log Z^{(j_0)}(\beta) + \beta y]$: the Legendre envelope is nondecreasing in y because each affine function in its infimum has slope $\beta > 0$. Its positivity at $x_1 > 0$ follows from the positive-energy states of the vacuum character and the ground term. Chernoff and the number-of-sectors bound give the following estimate in *natural logarithms*; the factor $\log 2$ converts the binary encoding parameter λ to natural-log units:

$$\lambda \log 2 \le \log \dim V_{\le h} \le mb + c\log(m+1). \tag{4.8}$$

Choose a maximizing sector σ^*. It cannot have ground energy $g_{\sigma^*} > mx$, since then $\mathcal{L}_{\sigma^*}(x) = -\infty$. Hence $2x - g_{\sigma^*}/m \ge x \ge x_1$, so Lemma 4.4 applied at energy $2h$ gives

$$\log \dim V_{\le 2h} \ge m\mathcal{L}_{\sigma^*}(2x) - C\sqrt{m} \ge \sqrt{2}\, mb - C\sqrt{m},$$

where the last step is Lemma 4.2. Now Lemma 2.4 and (4.8) yield

$$\rho_{\mathrm{win}} \ge \frac{\sqrt{2}\, mb - C\sqrt{m}}{2(mb + c\log(m+1))} \ge \frac{1}{\sqrt{2}} - O_{\mathcal{A},L,c,[x_1,x_2]}(m^{-1/2}),$$

using $b \ge b_0(x_1) > 0$. If the displayed numerator is negative, the inequality is trivial because $\rho_{\mathrm{win}} \ge 0$. □

4.4 Tensor powers

Theorem 4.6 (Tensor powers)**.** *If B is graded, $B_0 \neq 0$, its character is finite for $\beta > 0$ with infinitely many states, and $\beta \mapsto \beta \log Z_B(\beta)$ is nonincreasing, then for h/m in any fixed compact of $(0, \infty)$ and $2^\lambda \leq \dim(B^{\otimes m})_h$,*

$$\rho_{\mathrm{win}}(B^{\otimes m}; h, \lambda) \geq 1/\sqrt{2} - O(m^{-1/2}).$$

Proof. Apply Proposition 4.5 to the single-letter alphabet (Z_B), with one sector. □

4.5 Local refinement

Lemma 4.7 (Local scaling)**.** *For any $\varphi : (0, \infty) \to \mathbb{R}$, $\psi(\beta) = \beta\varphi(\beta)$, $\mathcal{L}(y) = \inf_{\beta>0}[\varphi(\beta) + \beta y]$, and $\beta > 0$,*

$$\varphi(\beta) + 2y\beta \geq \sqrt{2}\,\mathcal{L}(y) + \frac{\psi(\beta) - \psi(\sqrt{2}\beta)}{\beta}. \tag{4.9}$$

Proof. Subtract $\sqrt{2}[\varphi(\sqrt{2}\beta) + \sqrt{2}\beta y]$ from the left side. The difference is $\varphi(\beta) - \sqrt{2}\varphi(\sqrt{2}\beta) = [\psi(\beta) - \psi(\sqrt{2}\beta)]/\beta$; the bracketed quantity is at least $\sqrt{2}\,\mathcal{L}(y)$. □

5 Free fields

Proposition 5.1 (Bosonic Fock spaces)**.** *Let $c_n \geq 0$ be nondecreasing nonnegative integers, not all zero, with $\sum_{n\geq 1} n c_n e^{-\beta n} < \infty$ for $\beta > 0$. For $Z(\beta) = \prod_{n\geq 1}(1 - e^{-\beta n})^{-c_n}$, the function $\beta \log Z(\beta)$ is strictly decreasing.*

Proof. Writing $G_c(q) = \sum_{n\geq 1} c_n q^n$ for the mode-multiplicity series and expanding the logarithm, we have $\beta \log Z(\beta) = \sum_{k\geq 1} k^{-2}\Phi(k\beta)$ with $\Phi(u) = uG_c(e^{-u})$. If $d_j = c_j - c_{j-1} \geq 0$ ($c_0 = 0$), then

$$\Phi(u) = \sum_{j\geq 1} d_j e^{-(j-1)u} \frac{u}{e^u - 1}.$$

Every summand decreases, and at least one strictly decreases. Hence so do Φ and the original series. □

This includes a single free boson ($c_n = 1$) and plane-partition-type multiplicities ($c_n = n$).

Proposition 5.2 (Neveu–Schwarz fermion)**.** *For $Z_F(\beta) = \prod_{n\geq 1}(1 + e^{-\beta(n-1/2)})$, the function $\beta \log Z_F(\beta)$ is strictly decreasing.*

Proof. Expanding $\log(1 + x)$ yields

$$\beta \log Z_F(\beta) = \sum_{k\geq 1} \frac{(-1)^{k+1}}{k^2}\, g(k\beta), \quad g(u) = \frac{u}{2\sinh(u/2)} = H(u^2/4), \quad H(s) = \prod_{n\geq 1}\Big(1 + \frac{s}{n^2\pi^2}\Big)^{-1}.$$

The product implies $b(s) := -H'(s) > 0$ is strictly decreasing to 0. Thus differentiating termwise on compact positive β-intervals gives

$$(\beta \log Z_F)'(\beta) = -\frac{\beta}{2} \sum_{k\geq 1} (-1)^{k+1} b(k^2\beta^2/4) < 0$$

by the alternating-series test. □

Corollary 5.3. *Finite tensor products of the Fock spaces in Propositions 5.1–5.2 meet the hypotheses of Theorem 4.6.*

Remark 5.4 (Finite blocks). If $1 < \dim B < \infty$ and $B_0 \neq 0$, then $(\beta \log Z_B)' \to \log \dim B > 0$ as $\beta \downarrow 0$. This violates the criterion but does not create admissible pieces with improved ρ.

6 Kleinian code lattices

6.1 Codes, lattices and characters

Put $\mathbb{K} = \mathbb{F}_2^2$, the Klein four-group, and endow it with the alternating form

$$\omega((a,b),(a',b')) = ab' + a'b.$$

For $S \subset \mathbb{K}^m$ an additive isotropic ($\mathbb{F}_2$-linear and $\omega|_S = 0$) code (as in the stabilizer-code framework [10]), define $\mathrm{wt}(s) = |\{i : s_i \neq 0\}|$. We *define* the graded vector space V_S, with degrees in $\frac{1}{2}\mathbb{Z}_{\geq 0}$, by prescribing the Hilbert series of its codeword sectors: each $s \in S$ contributes $A^{m-\mathrm{wt}(s)}B^{\mathrm{wt}(s)}$, with the series A, B specified in (6.1). Their coefficients are nonnegative integers (Lemma 6.1), so this character prescribes a graded vector space without requiring a vertex-algebra construction. The discriminant group $D_4^*/D_4 \cong \mathbb{K}$ carries the bilinear form $\omega/2 \pmod{\mathbb{Z}}$; its three nonzero cosets have identical theta series.

Optional lattice interpretation. The preimage Λ_S of S between D_4^m and $(D_4^*)^m$ is an integral (not necessarily even) lattice. Standard lattice-character formulas realize the same Hilbert series through a lattice vertex (super)algebra [16, 12], but no such algebraic structure is needed for the dimension estimates below. Related stabilizer-code/CFT constructions [13] concern a different geometric setting and are not invoked in the proofs. In particular, no Appendix-B-compatible operators are claimed on V_S.

Set $q = e^{-\beta}$, $P(q) = \prod_{n\geq 1}(1-q^n)$ and

$$\theta_3 = \sum_{n\in\mathbb{Z}} q^{n^2/2}, \quad \theta_4 = \sum_{n\in\mathbb{Z}} (-1)^n q^{n^2/2}, \quad \theta_2 = \sum_{n\in\mathbb{Z}} q^{(n+1/2)^2/2}.$$

The classical Jacobi quartic identity $\theta_3^4 = \theta_2^4 + \theta_4^4$ gives the equivalent forms of B below. In our normalization, the D_4 theta series begins

$$\Theta_{D_4}(q) = \frac{\theta_3^4 + \theta_4^4}{2} = 1 + 24q + 24q^2 + 96q^3 + \cdots.$$

The vacuum and the three equivalent nonzero-coset characters are

$$A(\beta) = \frac{\theta_3^4 + \theta_4^4}{2P^4}, \qquad B(\beta) = \frac{\theta_2^4}{2P^4} = \frac{\theta_3^4 - \theta_4^4}{2P^4}, \quad Z_{V_S} = \sum_{s\in S} A^{m-\mathrm{wt}(s)}B^{\mathrm{wt}(s)}. \tag{6.1}$$

Lemma 6.1 (Positive coefficients)**.** *$A = 1 + 28q + 134q^2 + 568q^3 + \cdots$, and $B = q^{1/2}\mathcal{C}(q)$ with $\mathcal{C}(q) = 8 + 64q + 288q^2 + 1024q^3 + \cdots$. Both $A - 1$ and $\mathcal{C}$ have nonnegative integral coefficients. In particular A, $-A'$ and $\mathcal{C}(e^{-\beta})$ are nonincreasing in β.*

Proof. The numerator is the theta series of a D_4 coset and P^{-4} is a product of geometric series. Coefficient positivity and the first coefficients follow by expansion. □

Lemma 6.2 (Decreasing ratio)**.** *$t := B/A \in (0,1)$ is strictly decreasing in β, and $v := (\beta \log t)' < 0$.*

Proof. The Jacobi triple product gives

$$r = \left(\frac{\theta_4}{\theta_3}\right)^4 = \prod_{n\geq 1}\left(\frac{1-q^{n-1/2}}{1+q^{n-1/2}}\right)^8, \qquad t = \frac{1-r}{1+r}.$$

Each factor of r increases with β, whence $t' < 0$ and $v = \log t + \beta t'/t < 0$. □

6.2 Sectors and weight bounds

Write $a_{m,k} = |\{s \in S : \mathrm{wt}(s) = k\}|$. Grouping characters gives

$$V_S = \bigoplus_{k : a_{m,k} > 0} M_k, \qquad Z_{M_k} = a_{m,k} A^{m-k} B^k, \tag{6.2}$$

and the ground energy of the sector is $k/2$. The vacuum sector M_0 has character A^m with multiplicity exactly $a_{m,0} = 1$, because the only codeword of weight zero is 0.

Lemma 6.3 (Weight bounds)**.** *For every isotropic additive $S \subset \mathbb{K}^m$,*

$$a_{m,k} \le 2^m, \qquad a_{m,k} \le \binom{m}{k} 2^k; \qquad \frac{1}{m} \log a_{m,k} \le \min\{\log 2,\ H(\kappa) + \kappa \log 2\}, \tag{6.3}$$

where $\kappa = k/m$ and $H(\kappa) = -\kappa \log \kappa - (1-\kappa)\log(1-\kappa)$.

Proof. S is isotropic in a symplectic space of dimension $2m$, so $\dim_{\mathbb{F}_2} S \le m$. For any fixed k-subset of coordinates T, words supported in T form an isotropic subspace of $\mathbb{K}^T$, of size at most 2^k. Finally $\binom{m}{k} \le e^{mH(k/m)}$. The additivity of S is essential. □

6.3 Sectorwise criterion

Define $\varphi_{m,k} = m^{-1} \log Z_{M_k}$ and

$$g(\beta) = (\beta \log A)', \quad F_1 = g + \log 2, \quad F_2 = g + \log(1 + 2e^v), \quad F = \min\{F_1, F_2\}. \tag{6.4}$$

Theorem 6.4 (Sectorwise criterion)**.** *For every $m \ge 1$, additive isotropic code $S \subset \mathbb{K}^m$, and present sector k, $\beta \mapsto \beta\varphi_{m,k}(\beta)$ is strictly decreasing on $(0, \infty)$.*

Proof. By (6.2) and $\log B = \log A + \log t$,

$$(\beta\varphi_{m,k})' = \frac{1}{m} \log a_{m,k} + g + \kappa v.$$

The first inequality in (6.3), together with $v < 0$, bounds this by F_1; the second gives

$$(\beta\varphi_{m,k})' \le g + \sup_{0 \le \kappa \le 1} \{H(\kappa) + \kappa(\log 2 + v)\} = g + \log(1 + 2e^v) = F_2.$$

Thus it is at most $F(\beta) < 0$ by Proposition A.1. No weight-spectrum limit or concavity is used. □

6.4 Main theorem and transfer

Here an *admissible level* for V_S is a positive degree $h \in \frac{1}{2}\mathbb{Z}_{>0}$ of its grading with $\dim(V_S)_h > 0$; for the source's integer-degree comparison we additionally require $h \in \mathbb{Z}_{>0}$. The constants in the next theorem depend on the density endpoints x_1, x_2 and the fixed alphabet (A, B), not on m, h, λ or the code S.

Theorem 6.5 (Strict fixed-density window barrier)**.** *Fix $\mathcal{J} = [x_1, x_2] \Subset (0, \infty)$. There are constants $\eta_{\mathcal{J}} > 0$ and $C_{\mathcal{J}} > 0$, independent of m, the additive isotropic code $S \subset \mathbb{K}^m$, the admissible level h, and $\lambda \ge 1$, such that, whenever $h/m \in \mathcal{J}$ and $2^\lambda \le \dim(V_S)_h$,*

$$\rho_{\mathrm{win}}(V_S; h, \lambda) \ge \frac{1}{\sqrt{2}} + \eta_{\mathcal{J}} - \frac{C_{\mathcal{J}}}{\sqrt{m}}. \tag{6.5}$$

In particular, for each fixed $\mathcal{J}$, the prescribed window cost is strictly greater than $1/\sqrt{2}$ for all sufficiently large m. The dependence on $\mathcal{J}$ is explicit in the hypotheses.

Proof. Write $x = h/m$ and let M_k denote the sectors in (6.2). Set $b = \max_{k : a_{m,k} > 0} \mathcal{L}_{M_k}(x)$ and choose a maximizing sector k^*. By the encoding assumption and the sector Chernoff bound,

$$\lambda \log 2 \le \log \dim(V_S)_h \le mb + \log(m+1). \tag{6.6}$$

For both single-letter characters A, B, choose positive $\beta_{\mathrm{lo}}(\mathcal{J}) < \beta_{\mathrm{hi}}(\mathcal{J})$ such that

$$\mu_j(\beta_{\mathrm{lo}}) > 2x_2, \qquad \mu_j(\beta_{\mathrm{hi}}) < g_j + x_1/4, \quad (j = A, B), \qquad g_A = 0,\ g_B = 1/2.$$

Existence follows from their positive excitation coefficients and the limits of their mean energies. These choices work uniformly for the maximizing sector, since its ground energy is at most x whenever its Legendre envelope at x is finite. Let $V_*(\mathcal{J})$ be a uniform bound on the relevant single-letter variances in this compact inverse-temperature interval. For all sufficiently large m, the constructive Chebyshev

argument in Lemma 4.4 (with target $y = 2x$ and gap $2x - k^*/(2m) \geq x_1$) selects an inverse temperature $\beta_1 \in [\beta_{\mathrm{lo}}, \beta_{\mathrm{hi}}]$ for which

$$\log \dim(V_S)_{\leq 2h} \geq m\big[\varphi_{m,k^*}(\beta_1) + 2x\beta_1\big] - C'_{\mathcal{J}}\sqrt{m}. \tag{6.7}$$

Here one uses the *explicit* bound (4.6), not merely the lower bound involving the infimum $\mathcal{L}_{M_{k^*}}(2x)$: this distinction is needed in the next step. Indeed, applying Lemma 4.7 at that same β_1 (and recalling $\mathcal{L}_{M_{k^*}}(x) = b$) gives

$$\varphi_{m,k^*}(\beta_1) + 2x\beta_1 \geq \sqrt{2}\, b + \frac{\psi_{m,k^*}(\beta_1) - \psi_{m,k^*}(\sqrt{2}\beta_1)}{\beta_1}.$$

The sectorwise sign criterion (Theorem 6.4) supplies a continuous function $F(\beta) < 0$, independent of m, k, S, such that $\psi'_{m,k}(\beta) \leq F(\beta)$. Hence the last summand is at least

$$\delta_{\mathcal{J}} := (\sqrt{2} - 1) \min_{\beta \in [\beta_{\mathrm{lo}}(\mathcal{J}), \sqrt{2}\beta_{\mathrm{hi}}(\mathcal{J})]} (-F(\beta)) > 0.$$

For all sectors, the Chernoff envelope also has the uniform upper bound

$$b \leq b_{\mathcal{J}}^{\max} := \log 2 + \max\{\log A(1), \log B(1)\} + x_2.$$

Combining the central-window inequality $\rho_{\mathrm{win}} \geq D(2h)/(2\lambda)$ with (6.6) and (6.7) gives

$$\rho_{\mathrm{win}} \geq \frac{m(\sqrt{2}\, b + \delta_{\mathcal{J}}) - C'_{\mathcal{J}}\sqrt{m}}{2mb + 2\log(m+1)} \geq \frac{1}{\sqrt{2}} + \frac{\delta_{\mathcal{J}}}{2b_{\mathcal{J}}^{\max}} - \frac{C_{\mathcal{J}}}{\sqrt{m}}.$$

In the last step, the positive vacuum-sector lower bound on b for $x \in \mathcal{J}$ controls the denominators, and $\log(m+1)/m = O(m^{-1/2})$; these bounds are independent of S. Set $\eta_{\mathcal{J}} = \delta_{\mathcal{J}}/(2b_{\mathcal{J}}^{\max}) > 0$. Increasing $C_{\mathcal{J}}$ absorbs the finitely many shorter lengths. This establishes the claimed fixed-compact estimate without an assumption on concavity or convergence of code weight spectra. □

Corollary 6.6 (Uniform fixed-density window barrier)**.** *Under the hypotheses of Theorem 6.5,*

$$\rho_{\mathrm{win}}(V_S; h, \lambda) \geq \frac{1}{\sqrt{2}} - \frac{C_{\mathcal{J}}}{\sqrt{m}}$$

for a constant independent of the code S and m.

Proof. Immediate from Theorem 6.5, since $\eta_{\mathcal{J}} > 0$. □

Corollary 6.7 (Uniform strict separation)**.** *Fix $\mathcal{J} \Subset (0, \infty)$. For all sufficiently large m (depending only on $\mathcal{J}$), every family as in Theorem 6.5 satisfies $\rho_{\mathrm{win}} \geq 1/\sqrt{2} + \eta_{\mathcal{J}}/2$, regardless of whether its code-weight spectra are concave or converge.*

Remark 6.8 (Whole-character estimate). For $0 \leq t \leq 1$, the classical support argument yields $W_S(1, t) := \sum_{s \in S} t^{\mathrm{wt}(s)} \leq (1+t)^m$. Convexity gives

$$m^{-1}(\beta \log Z_{V_S})' \leq g + \log(1 + e^v).$$

This alone does not control the microcanonical windows without extra hypotheses; the proof above proceeds sector by sector.

Lemma 6.9 (Even subcode and integer-degree part)**.** *For isotropic S, the map $s \mapsto \mathrm{wt}(s) \bmod 2$ is $\mathbb{F}_2$-linear. Its kernel S_0 is isotropic, all its weights are even, $[S : S_0] \leq 2$, and*

$$\bigoplus_{E \in \mathbb{Z}} (V_S)_E \cong V_{S_0}$$

as graded vector spaces.

Proof. For $Q : \mathbb{K} \to \mathbb{F}_2$, $Q(0) = 0$ and $Q(x) = 1$ for $x \neq 0$, direct inspection gives $Q(x+y) + Q(x) + Q(y) = \omega(x, y)$. Thus $\mathrm{wt}(s) \bmod 2 = Q^{\oplus m}(s)$ is linear on isotropic S. A sector of weight k has character $a_{m,k} q^{k/2} A^{m-k} \mathcal{C}^k$ with $A, \mathcal{C} \in \mathbb{Z}[[q]]$, so it has integer degrees precisely for even k. Their sum is the character and the sector sum of V_{S_0}. □

Remark 6.10 (Integer degrees and admissibility). For integer-energy windows as in the source Appendix B, apply Corollary 6.6 to S_0; it remains a code in the same class, with the same universal constants.

Lemma 6.11 (Growth at the exact integer level)**.** *Let $\mathcal{J} = [x_1, x_2] \Subset (0, \infty)$. There exist $c_{\mathcal{J}}, C_{\mathcal{J}} > 0$ and an integer $m_{\mathcal{J}}$ such that, for every $m \geq m_{\mathcal{J}}$ and every* integer *h with $h/m \in \mathcal{J}$, every isotropic S satisfies*

$$c_{\mathcal{J}} m \leq \log_2 \dim(V_S)_h \leq C_{\mathcal{J}} m. \tag{6.8}$$

Proof. The vacuum sector gives $\dim(V_S)_h \geq [q^h]A^m$. Write $A = N(q)P^{-4}$ with $N = (\theta_3^4 + \theta_4^4)/2 \in \mathbb{Z}_{\geq 0}[[q]]$ and constant term 1. Because $P^{-4m} = (1-q)^{-1}G(q)$ for some positive-coefficient G, its coefficients are nondecreasing; multiplication by N^m preserves this property. Therefore

$$[q^h]A^m \geq \frac{1}{h+1} \dim(A^m)_{\leq h}.$$

Apply Lemma 4.4 to the single-letter alphabet (A) with $y = h/m \in \mathcal{J}$ and gap $\delta = x_1$. It yields $\log \dim(A^m)_{\leq h} \geq m b_A(x_1) - C\sqrt{m}$ with $b_A(x_1) > 0$, hence $\log \dim(V_S)_h \geq m b_A(x_1) - C\sqrt{m} - \log(h+1)$. For example, set $b_* = b_A(x_1) > 0$ and choose $m_{\mathcal{J}}$ sufficiently large that the finite-length lemma applies and, for every $m \geq m_{\mathcal{J}}$,

$$C\sqrt{m} \leq \tfrac{1}{4} m b_*, \qquad \log(x_2 m + 1) \leq \tfrac{1}{4} m b_*.$$

Then one may take $c_{\mathcal{J}} = b_*/(2\log 2)$; this specifies a finite threshold $m_{\mathcal{J}}$ without claiming its numerical optimality. For the other direction, use (4.2) for each of the at most $m+1$ sectors at $\beta = 1$: $\log \dim(V_S)_h \leq \log(m+1) + m[\log 2 + \max\{\log A(1), \log B(1)\} + x_2]$. □

Corollary 6.12 (Transfer to Definition 7(b))**.** *Fix a compact density interval $\mathcal{J} \Subset (0, \infty)$, $\theta > 0$ and an integer $c \geq 1$. Assume:*

(i) *$S \subset \mathbb{K}^m$ is additive isotropic and $h/m \in \mathcal{J}$;*

(ii) *$\lambda \geq \theta m$ and $2^\lambda \leq \dim(V_S)_h$;*

(iii) *the piece index set I contains 0;*

(iv) *the two bit costs at each s are the* ceilings *of the binary logarithms of the prescribed window dimensions and obey (2.1).*

Then

$$\rho \geq \frac{1}{\sqrt{2}} - C(\mathcal{J}, \theta, c) m^{-1/2}. \tag{6.9}$$

For integer h one may choose $\lambda = \lfloor \log_2 \dim(V_S)_h \rfloor$, which obeys $\lambda = \Theta_{\mathcal{J}}(m)$ by Lemma 6.11. The conclusion applies equally to the integer-degree even subcode S_0.

Proof. At $s = 0$ the two costs are each at least $D(2h)$, and (2.1) gives

$$\rho \geq \frac{D(2h)}{2\lambda} - \frac{c \log(2\lambda)}{4\lambda}.$$

The *central-ratio* estimate in Proposition 4.5 gives the first term at least $1/\sqrt{2} - O_{\mathcal{J}}(m^{-1/2})$. From $\theta m \leq \lambda \leq C_{\mathcal{J}} m$, the tolerance is $O_{\mathcal{J},\theta,c}(\log m/m) = O(m^{-1/2})$. □

7 Limitations and open problems

Open directions. Beyond the setting of Section 1.4, the following questions concern varying density, other character alphabets and admissible operator constructions, in dialogue with the questions studied by [6].

Question 7.1 (Variable densities). Does Theorem 6.5 extend uniformly to $h/m \to 0$ or $h/m \to \infty$? The compact inverse-temperature bounds used in Lemma 4.4 degenerate in those regimes. The latter regime is especially relevant to the original partition construction.

Remark 7.2 (Heuristic only). Both ends may admit stronger specialized estimates; partition- and Cardy-type high-energy heuristics [11, 15] and vacuum-sector estimates at small density are suggestive, but they supply no uniform proof in m and S here.

Question 7.3 (Other glue groups). Can the sector criterion be established for other lattices and discriminant groups, with complete weight enumerators when the nonzero cosets are inequivalent? Proposition 4.5 would then apply without change.

Question 7.4 (Admissible graded operators). Do the operator relations required by [4, Appendix B] force a corresponding thermodynamic saturation condition, or can exact efficiently computable pieces be built on a family with a genuinely smaller window cost? Such operators are not constructed in this paper.

Question 7.5 (Intrinsic ranks of A_0: two distinct problems). *Fixed partition encoding:* for the central piece A_0 and the partition encoding *specified* by [4, Appendix B], determine the intrinsic row and column flattening ranks and whether they are at least $\dim V_{\le 2h}/\mathrm{poly}(h)$. *Uniform admissible encodings:* does an analogous asymptotic lower bound hold *uniformly over every admissible encoding*? These are different quantified statements. Neither follows from the present window estimates.

Question 7.6 (Universal floor). For a genuinely broader class of exact algebraic pieces, is there any $\varepsilon > 0$ that forces $\rho \ge 1/2 + \varepsilon$? This is well beyond the results of this note.

Use of AI

This research was conducted with substantial assistance from ChatGPT (OpenAI) and Claude (Anthropic). The models were used for mathematical exploration, candidate arguments, drafting, program development and critical discussion. The thermodynamic saturation criterion and sectorwise approach were developed in this AI-assisted, author-directed process; an exclusive individual attribution to either model is not asserted. The author selected the research question, reviewed and organized the arguments, and assumes full responsibility for the mathematical statements, software and citations. No AI system is an author.

Computational verification

The three numerical sign checks used in Appendix A were executed with Arb interval arithmetic and cross-checked in `mpmath.iv`. An author-run local Arb certificate is recorded in the public repository described in Appendix B; the calculations validate the specified numerical inequalities, conditional on the analytical reductions in the text. They do not constitute proof-assistant formalization of the full theorem. The manuscript has not undergone independent qualified human peer review.

Code availability

The programs, execution logs, pinned dependencies and checksums are publicly available in the versioned GitHub release identified in Appendix B. The release version labels the reproducibility software, not the scientific version of this manuscript.

A Proof of the sector sign

Proposition A.1 (Global sign). *For F in (6.4), $F(\beta) < 0$ for every $\beta > 0$. Specifically $F_1 < 0$ on $(0, 4]$ and $F_2 < 0$ on $[4, \infty)$.*

The sign proof uses the three regimes $(0, 4]$, $[4, 5]$, and $[5, \infty)$. The lemmas below establish the required inequalities on these intervals; their numerical enclosures can be checked with the companion Arb script.

Lemma A.2 (Modular transformation). *Put $\beta^* = 4\pi^2/\beta$, $\Psi(q) = (\theta_3(q)^4 + \theta_2(q)^4)/P(q)^4$, and $\psi_\Psi(y) = \log \Psi(e^{-y})$. Then*

$$\log A(\beta) = \frac{\beta^* - \beta}{6} - \log 2 + \psi_\Psi(\beta^*), \quad F_1(\beta) = -\frac{\beta}{3} + \psi_\Psi(\beta^*) - \beta^* \psi'_\Psi(\beta^*). \tag{A.1}$$

Moreover, by the Jacobi identity, $\Psi = A + 3B$, with the explicit initial expansion

$$\Psi(q) = 1 + 24q^{1/2} + 28q + 192q^{3/2} + 134q^2 + O(q^{5/2}).$$

Thus $\Psi(e^{-y}) = 1 + \sum_{\nu \in \frac{1}{2}\mathbb{Z}, \nu \geq 1/2} c_\nu e^{-\nu y}$ *with* $c_\nu \geq 0$. *These coefficients permit an independent low-order check of the point bound* $R(\pi^2) \approx 10.2925020821$, *while its rigorous enclosure is supplied by the certificate.*

Proof. The classical modular transformations and Poisson summation (see, e.g., [9]) give

$$\theta_3(e^{-\beta}) = (2\pi/\beta)^{1/2}\theta_3(e^{-\beta^*}),$$
$$\theta_4(e^{-\beta}) = (2\pi/\beta)^{1/2}\theta_2(e^{-\beta^*}),$$
$$P(e^{-\beta}) = (2\pi/\beta)^{1/2}e^{(\beta-\beta^*)/24}P(e^{-\beta^*}).$$

Substitution and differentiation using $\beta\beta^* = 4\pi^2$ give (A.1). Positivity follows from the theta and P^{-4} series. □

Lemma A.3 (Small inverse temperature)**.** $F_1(\beta) < 0$ *for* $0 < \beta \leq 4$.

Proof. Let $T_\Psi(y) := -\mathrm{d}\Psi(e^{-y})/\mathrm{d}y = \sum_\nu \nu c_\nu e^{-\nu y} > 0$. Since all $\nu \geq 1/2$ and $\Psi \geq 1$, $\psi_\Psi \leq \Psi - 1 \leq 2T_\Psi$ and $-\psi_\Psi' \leq T_\Psi$. Thus

$$F_1(\beta) \leq -\frac{\beta}{3} + (2+\beta^*)T_\Psi(\beta^*) = \frac{R(\beta^*) - 4\pi^2/3}{\beta^*}, \quad R(y) = y(2+y)T_\Psi(y).$$

Each term of R decreases for $y > 1 + \sqrt{5}$; for $\beta \leq 4$ we have $\beta^* \geq \pi^2$. The certified point bound is $R(\pi^2) < 10.29251 < 4\pi^2/3 \approx 13.15947$. Hence the right-hand side is negative. □

Lemma A.4 (Intermediate boxes)**.** $F_2(\beta) < 0$ *for* $4 \leq \beta \leq 5$.

Proof. Let $\Delta_A(\beta) = -A'(\beta) \geq 0$ and write $B = q^{1/2}\mathcal{C}$. Then

$$g = \log A - \frac{\beta\Delta_A}{A}, \quad v \leq -\beta + \log\mathcal{C} - \log A + \frac{\beta\Delta_A}{A}.$$

Since $\frac{\mathrm{d}}{\mathrm{d}\beta}\mathcal{C}(e^{-\beta}) \leq 0$ and $A \geq 1$, the preceding estimate also gives the bound required in the final regime:

$$e^{v(\beta)} \leq e^{-\beta}\mathcal{C}(e^{-\beta})\exp\big(\beta\Delta_A(\beta)\big). \tag{A.2}$$

For $4 \leq b' \leq \beta \leq b'' \leq 5$, coefficient positivity and the decrease of $A, \Delta_A, \mathcal{C}$ imply

$$F_2(\beta) \leq U(b', b'') := \log A(b') - \frac{b'\Delta_A(b'')}{A(b')} + \log\left(1 + \frac{2e^{-b'}\mathcal{C}(e^{-b'})}{A(b'')}e^{b''\Delta_A(b')/A(b'')}\right). \tag{A.3}$$

The ten intervals $[4+i/10, 4+(i+1)/10]$ have strictly negative upper bounds U, with the weakest no larger than -0.1618. Table 2 lists values generated with `mpmath.iv`; Arb independently verifies the negativity of all ten box bounds, with the largest upper endpoint below -0.16184 (Appendix B). □

Lemma A.5 (Large inverse temperature)**.** $F_2(\beta) < 0$ *for* $\beta \geq 5$.

Proof. Write $\varepsilon = A - 1 \geq 28q$, $\Delta_A = -A' \geq \varepsilon$. Since $\log(1+\varepsilon) \leq \varepsilon$,

$$g \leq \frac{\varepsilon(1 + \varepsilon - \beta)}{1 + \varepsilon}.$$

Furthermore (A.2) gives $e^v \leq q\mathcal{C}(e^{-\beta})\exp(\beta\Delta_A)$; also $\log(1 + 2e^v) \leq 2e^v$. The coefficients of $A - 1$ occur in degrees $n \geq 1$, so

$$\frac{\mathrm{d}}{\mathrm{d}\beta}(\beta\Delta_A(\beta)) = \sum_{n\geq 1} n a_n (1 - \beta n)e^{-\beta n} \leq 0 \qquad (\beta \geq 5).$$

Therefore ε, $\mathcal{C}$, and $\beta\Delta_A$ are all nonincreasing on $[5, \infty)$, and

$$F_2(\beta) \leq -e^{-\beta}\left[\frac{28(\beta - 1 - \varepsilon(5))}{1 + \varepsilon(5)} - 2\mathcal{C}(e^{-5})e^{5\Delta_A(5)}\right].$$

The bracket increases with β; at $\beta = 5$ certified enclosures give $89.162296\ldots - 46.224428\ldots > 42.9378 > 0$. □

Proof of Proposition A.1. Lemma A.3 gives $F(\beta) \leq F_1(\beta) < 0$ up to 4; Lemmas A.4 and A.5 give $F(\beta) \leq F_2(\beta) < 0$ above 4. □

β	$A(\beta)$	$\Delta_A(\beta)$	$\mathcal{C}(e^{-\beta})$	$U[\beta,\beta+.1]$ upper bound
4.0	1.5614948801	0.6140840464	9.2754793271	-0.1618
4.1	1.5035672897	0.5459791406	9.1446615704	-0.2186
4.2	1.4520193458	0.4862735504	9.0280978931	-0.2654
4.3	1.4060725724	0.4337747160	8.9240792320	-0.3029
4.4	1.3650575244	0.3874875235	8.8311300163	-0.3318
4.5	1.3283959991	0.3465767090	8.7479711845	-0.3528
4.6	1.2955865877	0.3103372854	8.6734899373	-0.3669
4.7	1.2661928643	0.2781711162	8.6067148458	-0.3750
4.8	1.2398336691	0.2495682349	8.5467952494	-0.3778
4.9	1.2161750683	0.2240918518	8.4929841075	-0.3763
5.0	1.1949236629	0.2013662464	8.4446236514	–

Table 2: Values for the sign check on $[4,5]$. The first three data columns are 10-decimal approximations to narrow interval enclosures; the last column gives outward-rounded upper bounds (four decimals) for U on ten subintervals. The table is computed with `mpmath.iv`; Arb independently verifies the ten-box maximum, not each printed row.

B Reproducibility

The numerical signs used in Proposition A.1 can be recomputed with four companion scripts: the canonical `certify_arb_independent.py`, the exact coefficient checker `series_exact.py`, the interval cross-check `certify_mpmath_iv_crosscheck.py`, and `regenerate_interval_table.py`. They check the small-, intermediate-, and large-inverse-temperature regimes and the illustrative eleven-row interval table. The interval arithmetic uses Arb [17] through `python-flint` 0.9.0 at 256-bit precision; the separate cross-check uses `mpmath` 1.3.0.

```
python -m pip install python-flint==0.9.0 mpmath==1.3.0
python certify_arb_independent.py
python supplement/series_exact.py
python supplement/certify_mpmath_iv_crosscheck.py
python supplement/regenerate_interval_table.py
```

The public versioned reproducibility archive is available at https://github.com/MFRADJ/graded-window-barrier-certificates/releases/tag/v1.0.0. The GitHub release is labelled `v1.0.0` (software and archive release); it accompanies the present manuscript, scientific version `v1.0`. The archive contains the four scripts, dependency specifications, recorded outputs, interval-table data and SHA-256 checksums. The typesetting source of the present editorial revision accompanies the preprint submission. In the distributed Arb program the box bound is labelled (A.3), consistent with equation (A.3) in the present text. The interval computations certify the implemented sign conditions subject to the stated analytical reductions; they do not constitute a formal proof of the entire theorem.

References

[1] R. Impagliazzo and R. Paturi, *On the complexity of k-SAT*, J. Comput. System Sci. 62 (2001), 367–375.

[2] R. Impagliazzo, R. Paturi, and F. Zane, *Which problems have strongly exponential complexity?*, J. Comput. System Sci. 63(4) (2001), 512–530. https://doi.org/10.1006/jcss.2001.1774.

[3] OpenAI, *The Exponential Time Hypothesis is False*, OpenAI Math Release preprint, revised 25 October 2026. https://github.com/openai/math/tree/main/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026.

[4] Claude (Anthropic), *A $2^{o(n)}$ time algorithm for 3-SAT*, Hexagon 2610.00577v1 (2026). https://hexagonmath.org/2610.00577v1.

[5] Carlos Stein Naves de Brito, *The partition exponent in the graded pieces of the Claude 3-SAT algorithm is optimal for pieces of that shape*, Hexagon 2610.00692v1, 31 October 2026. https://hexagonmath.org/2610.00692v1.

[6] Ruhollah Majdoddin, *Lowered crossover points of Anthropic's subexponential 3-SAT algorithm*, arXiv:2610.41622 (30 October 2026). https://arxiv.org/abs/2610.41622.

[7] Vaibhav Tulsyan, *What the ETH refutations do not give: barriers for the counting engines, with machine-checked lemmas*, Hexagon 2611.00149v1 (6 November 2026). https://hexagonmath.org/2611.00149v1.

[8] G. H. Hardy and S. Ramanujan, *Asymptotic formulae in combinatory analysis*, Proc. London Math. Soc. (2) 17 (1918), 75–115.

[9] T. M. Apostol, *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed., Springer (1990).

[10] A. R. Calderbank, E. M. Rains, P. W. Shor, N. J. A. Sloane, *Quantum error correction via codes over GF(4)*, IEEE Trans. Inform. Theory 44 (1998), 1369–1387.
[11] J. L. Cardy, *Operator content of two-dimensional conformally invariant theories*, Nucl. Phys. B 270 (1986), 186–204.
[12] J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer (1999).
[13] A. Dymarsky, A. Shapere, *Quantum stabilizer codes, lattices, and CFTs*, J. High Energy Phys. 2021(3), 160; arXiv:2009.01244.
[14] R. S. Ellis, *Entropy, Large Deviations, and Statistical Mechanics*, Springer (1985).
[15] T. Hartman, C. A. Keller, B. Stoica, *Universal spectrum of 2d conformal field theory in the large c limit*, J. High Energy Phys. 2014(9), 118; arXiv:1405.5137.
[16] G. Höhn, *Self-dual codes over the Kleinian four group*, Math. Ann. 327 (2003), 227–255; arXiv:math/0005266.
[17] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Computers 66 (2017), 1281–1292.